Paleomagnetic constraints on the tectonic history of the Foreland Belt, southern Canadian Cordillera: preliminary results[1]

R.J. Enkin, K.G. Osadetz, P.M. Wheadon, and J. Baker

Abstract: In the southern Canadian Cordillera, the paleomagnetic memory of Paleozoic carbonate strata in the Front Ranges and Inner Foothills of the Foreland thrust and fold belt retains no record of their known deposition at low latitudes. Instead, each folded structure exhibits a similar, but asynchronous, sequence of events including an eastwardly progressing, predeformational chemical remagnetization during the Cordilleran orogeny. The remagnetization of a "western Front Ranges" structure occurs during a period of normal polarity before 130 Ma. The paleomagnetic pole requires that the subsequent deformation of the western Front Ranges is Jurassic or younger. The remagnetization of a Front Ranges structure in the Lewis thrust sheet occurs during a period of normal polarity after 130 Ma but before deformation which, from other evidence, occurred around 75 Ma. The predeformational remagnetization of an "Inner Foothills" structure occurs during a reversed magnetic period that we interpret to be after 75 Ma. An Early Cretaceous sill in the Lewis thrust sheet was remagnetized during a reverse-polarity chron prior to the end of Lewis thrust deformation, when about 70% of the present dip of the sill was acquired. Remagnetization consistently predates deformation, whereas it occurs later at more easterly localities. There are also similarities in character and style of the remagnetizations among localities. When coupled with the eastward progression of the deformation, our observations suggest that an important and pervasive, but hitherto unrecognized and unappreciated, orogenic chemical process affected Paleozoic carbonate strata in the van of the deforming Cordilleran tectonic wedge.

Résumé : Dans la Cordillère canadienne méridionale, la mémoire paléomagnétique des strates de carbonates paléozoïques, dans les chaînes frontales des Rocheuses et des Foothills intérieurs de chevauchements et plissements de l'Avant-pays, ne retient aucun registre de leur dépôt identifié aux basses latitudes. Au lieu de cela, chaque structure plissée exhibe une séquence d'évènements similaires, mais asynchrone, incluant une réaimantation chimique avant la phase de déformation et qui progessait vers l'est durant l'orogenèse de la Cordillère. La réaimantation d'une structure « chaînes frontales des Rocheuses occidentales » fut acquise durant une période de polarité normale avant 130 Ma. Le pôle paléomagnétique implique que la déformation subséquente dans la région occidentale des chaînes frontales des Rocheuses a eu lieu au Jurassique ou à une période plus récente. La réaimantation d'une structure chaînes frontales des Rocheuses dans l'écaille de chevauchement de Lewis fut acquise durant une période de polarité normale postérieure à 130 Ma, mais selon d'autres enseignements, ce serait avant la déformation apparue vers 75 Ma. La réaimantation avant déformation d'une structure « Foothills intérieurs » fut acquise durant une période d'aimantation inverse, que nous croyons être survenue postérieurement à 75 Ma. Un filon-couche du Crétacé précoce dans l'écaille de chevauchement de Lewis fut réaimanté durant un chron de polarité inverse avant la phase terminale de la déformation associée au chevauchement de Lewis, lorsque environ 70% du pendage actuel du filon-couche fut atteint. La réaimantation antidate de manière constante la déformation, tandis dans les localités plus à l'est elle se manifeste plus tardivement. Des similitudes dans le style et le caractère des réaimantations apparaissent parmi les localités. Nos observations amalgamées à la progression vers l'est de la déformation, suggèrent qu'un processus chimique orogénique majeur et pénétrant, jusqu'à présent méconnu et sous-estimé, a affecté les strates de carbonates paléozoïques à la tête du prisme tectonique qui déformait la Cordillère.
[Traduit par la rédaction]

Introduction

The Foreland Belt of the western Canadian Cordillera is a superb laboratory for the study of thrust and fold belts. Orogenesis has transformed the ancestral North American miogeoclinal succession into a crustal accretionary wedge at high elevation. Within that transformation lies a record of the response of the North American craton to the accretionary growth of the western margin of North America through the Mesozoic and Cenozoic. The deformed strata contain great natural resources of hydrocarbons, coal, and sulphur. The architecture of the Foreland Belt has come to be well known through extensive mapping and structural analysis (Price 1994). Yet the environment, mechanism, and timing of Cordilleran deformation remain incompletely described. In this contribution, we apply paleomagnetic methods to broaden the characterization of orogenic processes, while further constraining the timing of deformational events in the Foreland Belt.

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R.J. Enkin,[2] P.M. Wheadon, and J. Baker. Geological Survey of Canada, P.O. Box 6000, Sidney, BC V8L 4B2, Canada.
K.G. Osadetz. Geological Survey of Canada, 3303 – 33rd Street, Calgary, AB T2L 2A7, Canada.

[1] Geological Survey of Canada Contribution 1996281.
[2] Corresponding author (e-mail: enkin@pgc.NRCan.gc.ca).

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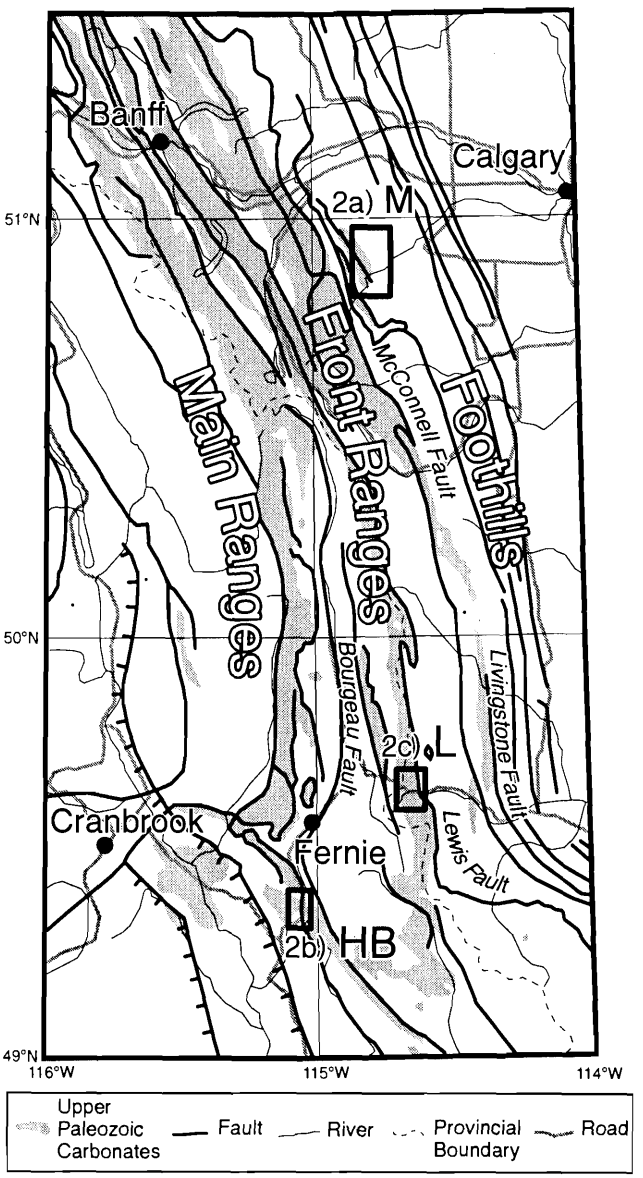
The Foreland Belt is a deformed, shortened, and thickened succession of platformal and miogeoclinal strata of Middle Proterozoic to Middle Jurassic age. They are overlain by westerly derived Columbian (Upper Jurassic – Lower Cretaceous) and Laramide (Upper Cretaceous – Paleogene) orogenic clastic wedges (Gabrielse and Yorath 1992). The sedimentary prism, originally deposited on the North American craton, has been deformed and accreted to the converging and impinging allochthonous terranes accreted to North America starting in the interval Middle Jurassic to Paleogene (Monger and Price 1979; Gabrielse and Yorath 1992). Crustal thickening accompanying accretion, and accompanied by thrusting and folding, builds topography, while flexing the continental lithosphere. This causes the subsidence of a foreland basin into which syntectonic sediments, cannibalized from the evolving and eroding foreland belt, accumulate (Price 1973; Beaumont 1981). The Foreland thrust and fold belt extends into the United States, where it is termed the Sevier thrust and fold belt (Jordan 1981; Mudge 1982).

The general progression of the deformation is from hinterland to foreland, from west to east. Foreland Belt structure is dominated by craton-verging listric thrust faults. Thrusts exploit detachments that are linked by intervening steeper and westward-facing ramps (Douglas 1950). Thrusts bifurcate upward from a westward-dipping basal décollement that lies near the base of the sedimentary succession, such that the underlying basement is not involved in the deformation affecting the sedimentary succession. The folding of structurally higher and more westerly thrust sheets accompanying the motion of underlying and more easterly thrust sheets is common. Thus, the major motion on more westerly thrust faults was largely complete prior to the major motion on underlying and more easterly thrusts (Dahlstrom 1970).

The deformation encroached upon and incorporated parts of the syntectonic clastic wedge. The minimum age of strata involved in the deformation, particularly below thrust faults, provides a maximum age for the last major motion of the overriding thrust sheet. Elsewhere, but not within our study area, there are syn- and postorogenic plutons cutting thrust faults that provide minimum age constraints on some of the deformation. Foreland Belt compressive deformation is inferred to end in the Early to Middle Eocene with the onset of crustal extension in the Omineca Belt (Carr et al. 1987). There is, however, a general lack of temporal constraints on the motion of individual thrust faults, and few truly proximal foreland basin sediments that might further constrain the age of the deformation (McMechan and Thompson 1993), as is the case in the United States (Jordan 1981).

The Foreland Belt is commonly divided into morphotectonic regions (Fig. 1). Throughout the southern Foreland Belt of Canada, there is a classical westward progression from Foothills to Front Ranges to Main Ranges that coincides with changing stratigraphic relationships at major thrust faults. Where the Foothills are predominantly composed of imbricate sheets of Mesozoic Foreland Basin strata, they are termed Outer Foothills. The Inner Foothills are marked by the appearance of Carboniferous strata in the hanging wall of thrust faults, as at both the Moose Mountain culmination and the Livingstone Range (Fig. 2*a*). The Front Ranges coincide with the appearance of Devonian and older strata in overthrust sheets above the McConnell and Lewis thrust faults. The Lewis thrust sheet, with a thick Middle Proterozoic succession, marks the leading edge of the Front Ranges in southern Alberta (Fig. 2*c*). The Hosmer Nappe and Broadwood Anticline lie west of a major synclinorium, the Fernie Basin, in the western Front Ranges (Fig. 2*b*).

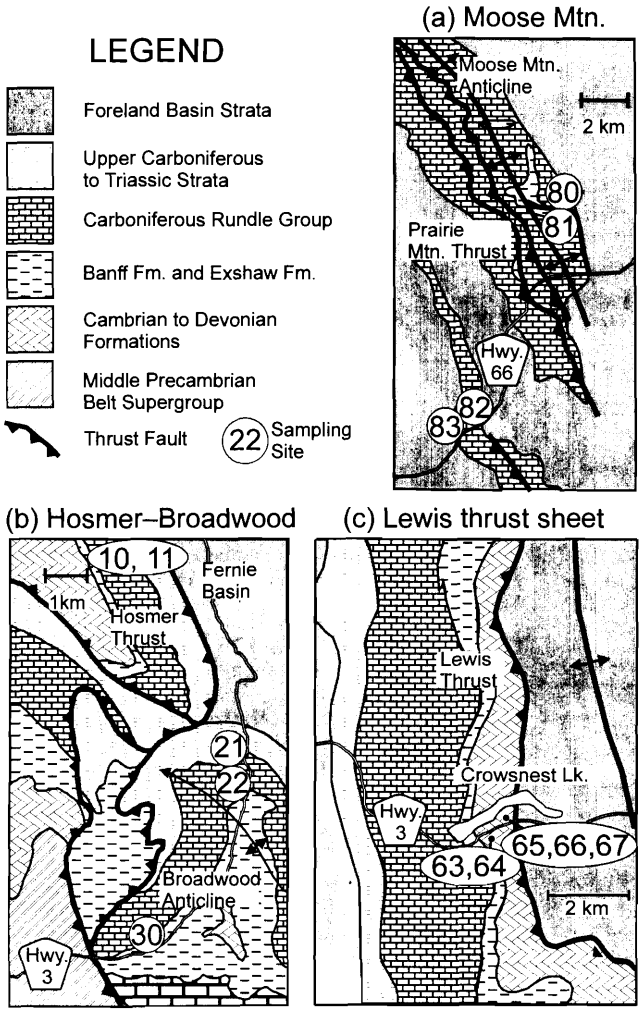
Fig. 1. General study setting between Calgary and Fernie, showing the morphotectonic belts of the Foreland Belt, select faults, and the outcrop pattern of upper Paleozoic carbonates. Approximate locations of more detailed sample locality maps (Figs. 2*a*–2*c*) are indicated. HB, Hosmer–Broadwood locality in the western Front Ranges; L, Lewis thrust sheet locality in the Front Ranges; M, Moose Mountain locality in the Inner Foothills.



Other work suggests that rocks in thrust and fold belts are particularly susceptible to remagnetizations (e.g., McCabe and Elmore 1989). Following reports of significant remagnetizations in the Cordilleran Foreland Belt (S. May, personal communication, 1993), we characterize and analyze these remagnetizations with the purpose of elaborating on the orogenic processes within, and the tectonic history of,

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Fig. 2. Sample locality maps. (*a*) The Moose Mountain culmination. (*b*) The Lizard segment of the Hosmer Nappe and Broadwood Anticline. (*c*) The Lewis thrust sheet at Crowsnest Lake. Specific sample sites are listed in Table 1.



the Cordilleran Foreland Belt. Most important is the timing of the remagnetization relative to the deformation. Thus the paleomagnetic fold test provides a useful tool for the analysis of orogenic remagnetizations, whereas the drift of the continents through time relative to the magnetic pole provides a clock that potentially dates the remagnetizations and constrains the motions on individual structures.

Sampling and analysis

This study reports (Fig. 1; Table 1) on a traverse through the Foreland Belt in southwestern Alberta and southeastern British Columbia. Samples were collected from folded or tilted beds to determine when the magnetic remanence was acquired relative to the local deformation. Samples for a fold test of the western Front Ranges were collected south of Fernie, British Columbia, from both the Lizard segment of the Hosmer Nappe, at the Snow Valley ski resort, and the Broadwood Anticline on either side of the Highway 3 road tunnel (Fig. 2*b*). The evolution of both structures is intimately related. The upright Broadwood Anticline plunges beneath the overturned limb of the Hosmer Nappe (Fig. 2*b*) (Benvenuto and Price 1979). A fold test at the leading edge of the Front Ranges was performed within the Lewis thrust sheet, near Crowsnest Lake, beside Highway 3 (Fig. 2*c*). There we sampled both a macroscopic fold, with a wavelength of approximately 10 m, and a Cretaceous trachyte sill and its contact zone. Samples for fold tests of Inner Foothills structures were collected from the Livingstone Range, both near Blairmore, Alberta, and in the Oldman River Gap, and from the Moose Mountain culmination in the vicinity of Canyon Creek and the Elbow Trail (Fig. 2*a*). Interpretable results were obtained from the Moose Mountain localities, but not from the Livingstone Range.

Most samples are from the Upper Devonian and Carboniferous carbonate succession. This succession includes Upper Devonian Fairholme Group, predominantly dolostone; Upper Devonian Palliser Formation, predominantly limestone; Upper Devonian and Lower Carboniferous Exshaw Formation, predominantly shale; Lower Carboniferous Banff Formation, predominantly lime mudstone and cherty lime mudstone; and Lower Carboniferous Rundle Group, predominantly lime grainstone in the lower part (Livingstone Formation), overlain by dolostones (Mount Head Formation) that are, in turn, overlain by arenaceous and silty dolostones with lesser green shale (Etherington Formation). Each locality consists of four or five sampling sites. Each site consists of 5–12 oriented cores (2.5 cm diameter) collected from a single formation. Sites span several metres to several tens of metres of stratigraphic section, in order to average secular variation and small-scale lithological effects.

Standard processing and measurements were applied to the collection. Specimens were cleaned using either thermal demagnetization with an ASC Scientific TD48 furnace or by alternating fields (AF) with a Schonstedt GSD-5 tumbling demagnetizer. All measurements were made on a Geofyzika JR5-A spinner magnetometer. Six to twelve demagnetization steps were applied to each specimen.

Paleomagnetic results

Classes of magnetic remanence characteristics

The remanence characteristics of sites divide clearly into three classes. Class A sites contain two components readily isolated using either thermal or alternating-field demagnetization techniques. Orthogonal component plots typically show two linear segments with a sharp corner at around 160°C or 10 mT treatment (Fig. 3). The first, the low-stability component (LSC), is sometimes quite scattered, whereas the second, the high-stability component (HSC), is almost always linear and decaying to the origin. The HSC in class A sites is usually tightly clustered (Fisher precision, $k > 50$). The natural remanent magnetization (NRM) intensity in class A sites is usually in the range 0.5–5 mA/m. Bulk susceptibilities (χ_0) tend to be greater than 5×10^{-6} SI/vol. Unblocking temperatures of NRM are quite low (<450°C), but the unblocking temperature of saturation isothermal remanence is closer to 550°C. These characteristics and the observation that magnetite is the predominant accessory mineral suggest that magnetite is the only remanence carrier.

Class B sites also contain LSC and HSC magnetizations, but their demagnetization spectra overlap. Often there was

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Table 1. Site information.

Site	Universal transverse Mercator coordinates	Formation	Age[a]	Bedding downdip azimuth (°)	Bedding dip (°)	No. of cores	NRM (mA/m)	χ_0 ($\times 10^{-6}$)	Q_N	Remanence class
Western Front Ranges locality: Hosmer Nappe — Broadwood Anticline (49.5°N, 115.1°W)										
10	6351E−54817N	Rocky Mountain	eP	37.2	110.8[b]	5	0.38	12.	0.88	A
11	6350E−54813N	Etherington	lM	35.8	110.5[b]	8	0.27	2.3	2.7	A
21	6442E−54684N	Etherington	lM	28.8	47.4	12	0.41	4.6	2.2	B
22	6441E−54684N	Etherington	lM	21.3	44.0	7	6.2	402.	0.39	A
30	6425E−54645N	Banff	eM	328.7	27.0	6	0.94	8.0	3.0	A
31	6413E−54640N	Livingstone	eM	274.3	30.1	12	0.10	−1.5	−2.1	C
Front Ranges locality: Lewis thrust sheet on Crowsnest Highway (49.6°N, 114.6°W)										
60	6669E−55000N	Mount Head	eM	260.7	45.6	17	0.50	24.	0.51	C
61	6678E−55002N	Livingstone	eM	252.6	30.6	8	0.32	3.9	2.1	C
63	6679E−55002N	Palliser	lD	247.7	36.6	6	2.0	24.	2.1	A
64	6680E−55002N	Palliser	lD	65.0	46.7	6	1.5	20.	1.9	A
65	6711E−55004N	Fairholme country rock	lD	246.9	56.1	6	3.8	19.	5.2	A
66	6711E−55004N	Crowsnest sill	98 ± 10 Ma	246.9	56.1	6	0.83	57.	0.36	A
67	6711E−55004N	Fairholme contact	lD	246.9	56.1	7	0.82	18.	1.1	A
Inner Foothills locality: Livingstone Range (50.1°N, 114.2°W)										
01	6884E−54856N	Misty	eP	88.2	50.5	6	0.38	13.	0.76	C
02	6997E−54277N	Misty	eP	272.7	33.1	6	0.67	43.	0.36	C
68	6854E−54986N	Etherington	lM	256.6	45.8	6	0.42	12.	0.92	C
70	6885E−55268N	Misty	eP	258.5	57.1	5	1.10	33.	0.61	C
71	6885E−55268N	Etherington	lM	259.7	52.0	6	0.30	27.	0.28	C
72	6886E−55268N	Etherington	lM	58.6	29.5	6	0.63	51.	0.31	C
73	6892E−55270N	Livingstone	eM	270.3	49.1	8	0.17	−3.2	−1.4	C
Inner Foothills locality: Moose Mountain (50.9°N, 114.8°W)										
80	6569E−56407N	Livingstone	eM	100.0	14.1	6	0.34	3.0	2.8	B
81	6572E−56405N	Livingstone	eM	111.1	15.8	6	0.19	−2.3	−2.1	B
82	6552E−56372N	Livingstone	eM	243.0	19.9	6	0.20	−2.9	−1.7	B
83	6551E−56360N	Livingstone	eM	230.1	20.8	6	0.58	34.	0.43	A

Notes: NRM, geometric mean of natural remanent magnetization intensity; χ_0, geometric mean of magnetic susceptibility in SI units; Q_N, geometric mean of Koenigsberger ratio.

[a]eP, Early Pennsylvanian; lM, Late Mississippian; eM, Early Mississippian; lD, Late Devonian.

[b]Overturned.

no component directed towards the origin (Fig. 3). It was usually possible to isolate the HSC of class B sites only by determining the intersection of great-circle demagnetization paths. The site mean LSC and HSC directions in class B sites tend to be similar to those of class A sites, although the results for class B sites tend to be more dispersed ($15 < k < 50$).

The HSC's of class C sites cannot be isolated. Typically, demagnetization paths start near their LSC direction, roughly in the present field direction. Upon progressive demagnetization the directions of class C sites tend towards random directions. Class C sites are usually quite noisy and cannot be interpreted. The results from class C sites are not reported in this paper.

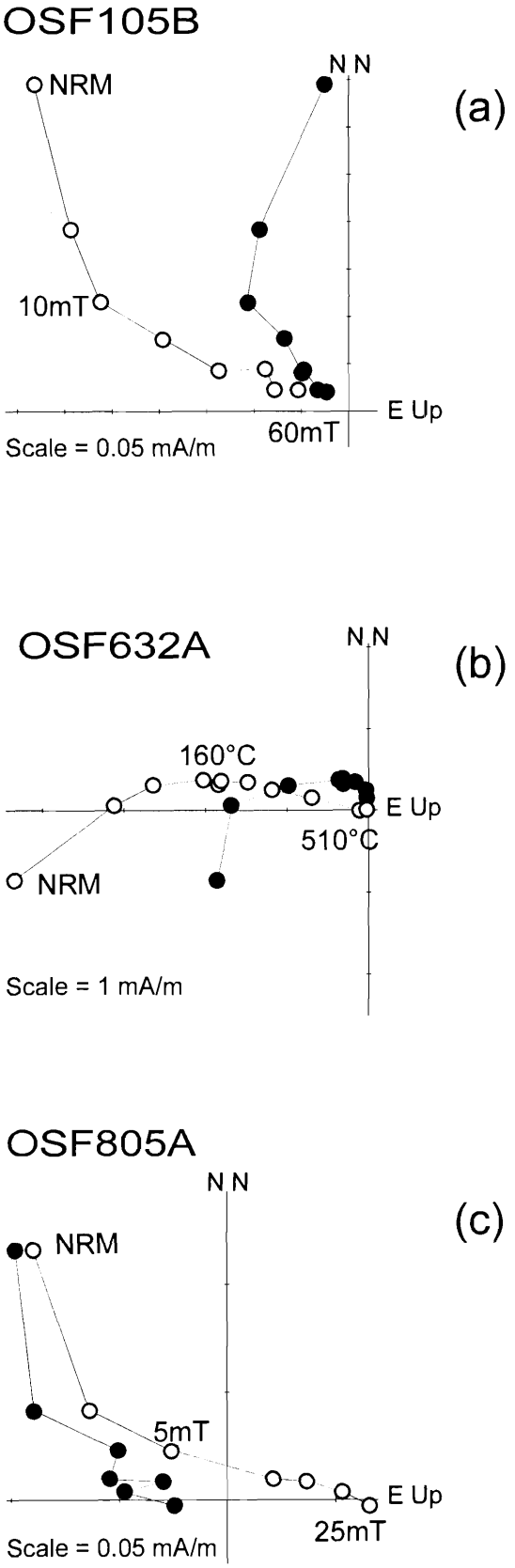
Among class B and class C sites, NRM and χ_0 values are indistinguishable, usually below 0.5 mA/m and 5×10^{-6} SI/vol., respectively (class C sites from the Livingstone Range have higher χ_0 but contain only a present field remanence). Often these sites are diamagnetic. This indicates that their susceptibility is dominated by contributions from the carbonate matrix over contributions from ferromagnetic carriers. Class A samples are inferred to be distinguished from class B and class C samples only by their abundance of magnetite. No other rock magnetic parameters, such as Koenigsberger ratio or median destructive temperature, appear associated with assigned class of remanence.

Fold-test results

The mean HSC directions of sites with class A and class B remanences are reported in Table 2. The directions are far better clustered after, rather than before, tectonic correction (Fig. 4). An unambiguously positive paleomagnetic fold test requires that maximum clustering occur on 100% untilting (Watson and Enkin 1993). Considering each of the three fold tests separately, we find that maximum clustering generally occurs at ~95% untilting (Table 3), but that the 95% confi-

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Fig. 3. Example orthogonal plots for (*a*, *b*) class A magnetization specimens, and (*c*) class B magnetization specimens. Steps given in milliTesla (mT) or degrees Celsius (°C). ●, projections onto the horizontal plane; ○, projections onto the vertical plane.



dence interval always includes 100% untilting. This indicates strongly that the stable remanence at each of the studied structures was acquired prior to deformation, while the strata were essentially flat lying.

When all the sites in this study are combined, we observe that maximum clustering occurs at significantly less than 100% untilting (92.2 ± 4.8%). This departure from complete unfolding indicates one of three possibilities. First, the predeformational remanence may include temporal variations in field direction among the set of structures studied. Alternatively, either the HSC was acquired during the earliest phases of the deformation, or there is a slight post-deformational overprint contaminating the predeformational remanence. We favour the first alternative, since mean directions are distinct and there is no mechanism to suggest that the remagnetization is contemporary through the three structures. Regardless, the origin of the HSC remanence predates the main stage of the local deformation.

Polarity of high stability component poles

Both polarities are represented in the collection. This provides both an indication of the stability of the remanence and proof that it is acquired asynchronously. Note that the magnetization polarity is uniform within individual sites and in groups of sites consituting a local fold test. The polarity of the predeformational HSC remanence of all specimens from sites constituting the Hosmer–Broadwood and Lewis thrust sheet localities is normal. The polarity of the same component for sites from the Moose Mountain locality is reversed.

High stability component pole directions

Pole positions (Fig. 5) provide clear evidence that the predeformational HSC remanence is not a primary, or depositional, remanence. Throughout the Devonian and Carboniferous, North America lay at low to subequatorial latitudes (e.g., Van der Voo 1990). Hence the geomagnetic inclination was consistently shallow at the time of sedimentation. Although there is no evidence of a shallow paleomagnetic inclination in any specimens from our collection, we still infer that such a remanence was once present. The observed magnetic inclinations from Devonian and Carboniferous carbonate formations are steep, mostly greater than 60°. North America did not lie at sufficiently high latitudes for these strata to acquire the observed steep magnetic inclinations prior to the Jurassic. Therefore, the predeformational HSC remanence was acquired in Jurassic and later times, during Cordilleran orogeny but prior to the local deformation.

Crowsnest sill data

Just above the Lewis Thrust, along the Crowsnest Highway, a 5 m wide trachyte porphyry sill intrudes Devonian Fairholme Group carbonates. It has been dated at ~98 Ma (K/Ar; D. Kisilevsky and D. Archibald, personal communication, 1994). Intrusion was contemporaneous with extrusion of the alkaline, Albian Crowsnest volcanics. The intrusive age and association indicate the sill was intruded during the Cretaceous Long Normal Superchron (CLNS). Carbonate country rock near the sill (site 65) exhibits the same predeformational, steep-inclination, normal-polarity remagnetization observed in the other Paleozoic carbonates sites

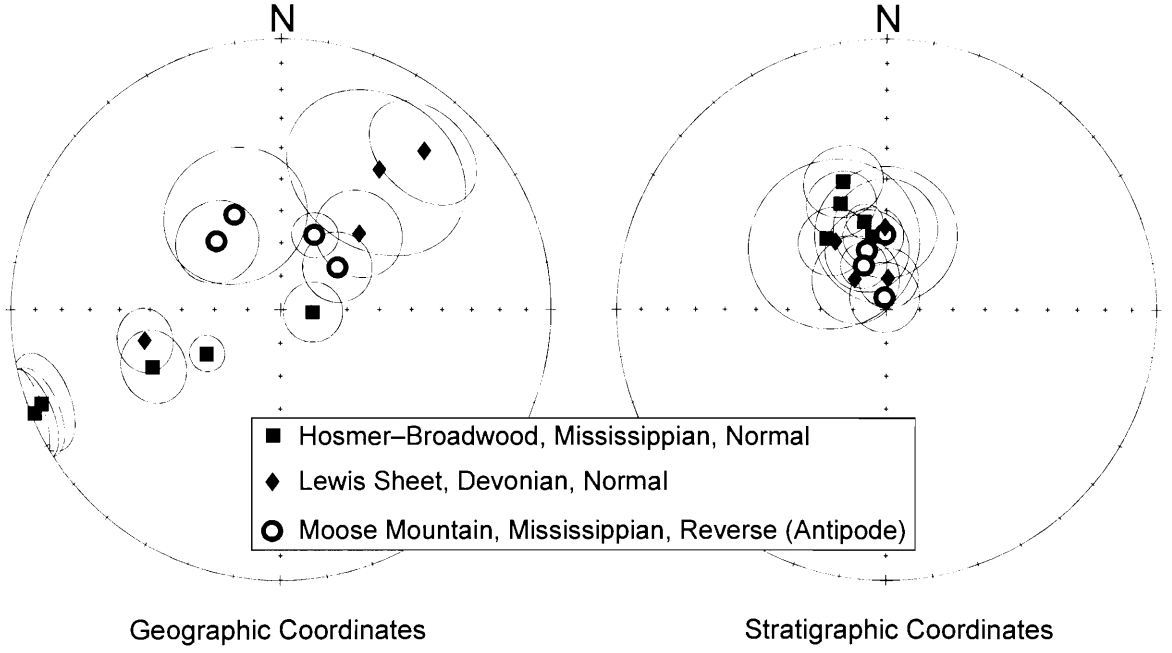
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Table 2. Paleomagnetic site means.

Site	n_D	n_C	D_G (°)	I_G (°)	D_S (°)	I_S (°)	k	a_{95} (°)	Remanence class	Comments
Western Front Ranges locality: Hosmer Nappe – Broadwood Anticline										
10	5	0	247.3	−1.4	336.7	54.8	57.2	10.2	A	Two distinct components, corner at 200°C, 20 mT
11	7	2	248.8	5.7	341.2	48.4	21.8	11.4	A	Two distinct components, corner at 200°C, 20 mT
21	6	2	245.9	47.1	321.6	63.6	30.1	10.4	B	Difficult to separate the two similar components
22	6	0	239.3	64.1	345.9	62.4	153.	5.4	A	HSC well defined from 100°C, 10 mT: LSC contaminated by HSC
30	5	1	94.4	80.3	349.6	67.4	57.9	9.0	A	Two distinct components, corner at 150°C, 10 mT
Front Ranges locality: Lewis thrust sheet at Crowsnest Lake										
63	4	0	45.7	56.8	313.8	77.0	49.2	13.2	A	Two distinct components, corner at 150°C, 10 mT
64	4	1	257.3	47.2	3.3	80.6	77.1	9.0	A	Two distinct components, corner at 150°C, 10 mT
65	3	0	42.2	22.0	359.2	65.2	65.0	15.4	A	*Country*: two distinct components, corner at 140°C, 10 mT
66	5	0	194.6	−65.2	95.4	−45.9	447.	3.6	A	*Intrusion*: two distinct components, corner at 200°C, 20 mT
67	4	0	34.9	36.8	321.3	63.9	13.6	25.8	A	*Contact*: two distinct components, corner at 150°C, 10 mT
Inner Foothills locality: Moose Mountain										
80	1	5	136.6	−61.7	161.5	−71.3	40.8	12.5	B	HSC determined by great circle intersection; probably contaminated by LSC
81	0	5	153.9	−57.9	179.4	−67.2	28.9	21.1	B	Noisy; HSC determined by great circle intersection; probably contaminated
82	3	3	233.1	−68.7	171.0	−86.2	47.6	10.5	B	Noisy, but acceptable isolation of HSC
83	7	0	204.6	−65.4	152.5	−75.0	77.0	6.9	A	Two distinct components, corner at 250°C, 15 mT

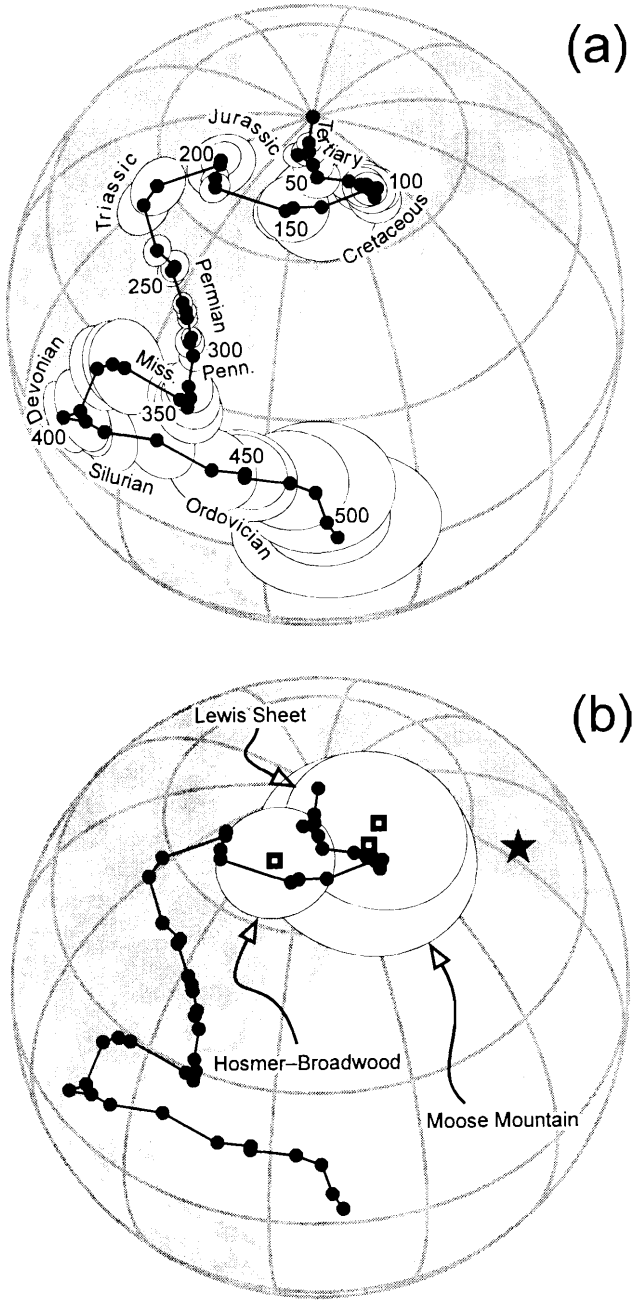
Notes: n_D, number of specimens rendering isolatable directions; n_C, number of great circle demagnetization paths; D_G and I_G, mean declination and inclination in geographic (uncorrected) coordinates; D_S and I_S, mean declination and inclination in stratigraphic (bedding corrected) coordinates; k, Fisher precision parameter; α_{95}, 95% confidence interval parameter. No high-stability component was isolated from class C sites.

Fig. 4. Site mean direction with associated 95% confidence circles before (in geographic coordinates) and after (stratigraphic coordinates) structural correction. The antipodes of reversed-polarity sites are illustrated. Each locality passes the fold test, indicating a predeformational magnetic remanence.



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Fig. 5. (*a*) North American apparent polar wander path with 95% confidence circles as compiled by Besse and Courtillot (1991) and Wynne et al. (1992). Ages in millions of years. (*b*) Mean pole with associated 95% confidence circle for each of our three localities. ★, sampling region. The pole positions indicate that the observed remanence is a Jurassic or younger overprint.



collected from the Lewis thrust sheet. The contact zone, between 5 cm and 2 m away from the sill (site 67), also has a predeformational pole and polarity similar to that observed in country rocks a greater distance from the sill. However, the sill (site 66) has a reversed HSC remanence and a paleomagnetic pole that is discordant with the North American apparent polar wander path (APWP) both before and after tectonic correction (Table 3).

Discussion

Acquisition of the predeformational remagnetization in Paleozoic carbonate strata

The data indicate that a similar sequence of events affects Paleozoic carbonate strata at all three fold-test localities. This sequence is (*i*) sedimentation, presumably with the acquisition of a low-latitude primary magnetic remanence, which is no longer preserved; (*ii*) complete remagnetization long after deposition, when North America had reached higher latitudes, during Cordilleran orogeny, either while the strata are flat, or nearly flat lying; and (*iii*) deformation of the remagnetized strata and the passive rotation of the remagnetized remanence during local deformation.

Pole positions (Fig. 5*b*) suggest the overprinting of depositional remanence occurs at different times among the three localities. Unfortunately, the age of the overprint cannot be precisely resolved using the paleomagnetic directions alone. There are two reasons for this lack of resolution. The first cause is intrinsic to the combination of the location of the study and the age of the remagnetization. The rate of apparent polar wander for North America during the interval Jurassic to Paleogene was quite moderate. The APWP has a poor geometry for resolving ages using pole positions, particularly during the Cretaceous standstill (90 ± 20 Ma, Fig. 5*a*). The second cause is the large uncertainty associated with each pole. The large uncertainties are attributable to the relatively small number of sites (4 or 5) collected at each locality.

The predeformational remagnetization direction determined for the Hosmer–Broadwood locality is either Jurassic to earliest Cretaceous, or Tertiary (Fig. 5*b*), when compared to the North American APWP (Besse and Courtillot 1991). The predeformational timing of the remagnetization and indications that structures of the western Front Ranges are Mesozoic (as outlined below) constrain the remagnetization to the older age range. Therefore, the determined pole position from the Hosmer–Broadwood fold test suggests a Jurassic to Neocomian age (200–130 Ma) for the predeformational remagnetization of Paleozoic carbonates involved in those structures. The mean value of the pole is close to 160 Ma (approx. early Late Jurassic).

The pole position from the positive fold test of a structure within the Lewis thrust sheet overlaps the APWP from Cretaceous (130 Ma) to present. The mean pole position determined for this fold test is closest to the Cretaceous standstill of the North American APWP (90 ± 20 Ma). The pole position from the Moose Mountain positive fold test is Upper Jurassic (150 Ma) to present. At Moose Mountain the mean pole position is similar to that determined for the Lewis thrust sheet positive fold test. The mean pole positions of all three localities fall close to the APWP, suggesting that the deformation of these structures has resulted in little vertical axis rotation.

The magnetic polarity within each fold-test locality is homogeneous. This implies that remagnetization events were short, perhaps less than one million years. The polarity of the remagnetization is normal at both the Hosmer–Broadwood and Lewis localities, but it is reversed at Moose Mountain. This proves that the timing of the remagnetization is different at different localities. From the pole positions we see that the

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Table 3. Paleomagnetic locality and global means.

Structure	N	0% untilt D (°)	0% untilt I (°)	0% untilt k	0% untilt α_{95} (°)	100% untilt D (°)	100% untilt I (°)	100% untilt k	100% untilt α_{95} (°)	Maximum clustering (% untilt)	VGP Lat. (°N)	VGP Long. (°E)	VGP dp (°)	VGP dm (°)
Hosmer–Broadwood	5	244.8	42.3	4.0	44.	338.9	59.6	78.	8.7	94.9 ± 8.5	72.6	131.5	9.8	13.1
Lewis thrust sheet	4	27.2	55.1	3.3	60.	338.1	72.8	53.	13.	96.1 ± 8.5	74.6	197.8	20.	22.8
Moose Mountain	4	176.1	−68.3	19.	22.	166.6	−75.2	85.	10.0	93.0 ± 23.6	76.6	217.3	17.	18.3
All	13	311.5	73.4	3.1	28.	340.4	68.5	49.	6.0	92.2 ± 4.8	77.4	172.8	8.6	10.1

Notes: *N*, number of sites; *D*, *I*, *k*, and α_{95}, Fisher average declination, inclination, precision factor, and 95% confidence interval; VGP, virtual geomagnetic pole; dp and dm, principal semiaxes of 95% confidence interval about pole. Note that site 66 is not included in the Lewis thrust sheet mean.

oldest remagnetization occurs in the westernmost locality. Therefore, we hypothesize that the remagnetization events were of short duration and that they occurred sequentially from west to east.

Pole positions show that the remagnetizations were all acquired contemporaneous with the deposition of the orogenic clastic wedges. Positive fold tests show that the remagnetizations were acquired prior to the local deformation at each locality. Pole positions and polarities both show that the predeformational remagnetizations were asynchronous. Furthermore, the same data suggest that the process of predeformational remagnetization, like the deformation, migrates eastward with time. Therefore, we suggest that our data record a pervasive eastward-migrating remagnetization of the undeformed stratigraphic succession that anticipates the incorporation of that portion of the Paleozoic platform into the Cordilleran accretionary prism. We cannot specify a process responsible for these remagnetizations. Yet the initiation and termination of the remagnetization can be inferred to be intrinsically linked to the progress of the Cordilleran orogeny. The remagnetization is a harbinger of the impinging crustal accretionary wedge occurring in the van of the deformation front.

Acquisition of the syndeformational remagnetization in the Crowsnest sill

Pole positions show that Paleozoic strata near and in the contact zone of the Cretaceous trachyte sill have completely lost their primary Paleozoic magnetizations. Both the country rock and the contact zone of the sill have poles like those determined for the positive fold test in the Lewis thrust sheet (Table 2). Unfortunately, it is impossible to determine the age of the predeformational remagnetization of Paleozoic strata relative to the age of the intrusion of the sill. The sill was intruded during the Cretaceous standstill of the North American APWP and the CLNS. We can expect no discernable difference between the remanence direction of the contact zone and that of country rock, regardless of the sequence of intrusion relative to the regional remagnetization. This is due entirely to the geometry of the North American APWP during the Cretaceous standstill and the persistent normal polarity during the CLNS.

The paleomagnetic remanence of the sill is not primary. The sill has a reversed polarity despite its CLNS emplacement age. Further, the pole, determined relative to either bedding or present horizontal, is discordant with the APWP for North America. Figure 6 illustrates that the pole determined for the sill falls on the APWP when it is untilted 10–30%, that is, when it had acquired 70–90% of the current dip. We infer that the remagnetization of the sill is syntectonic, having occurred late in the deformation of the sill. Therefore, the reverse remagnetization of the sill postdates the CLNS and must have occurred during chron 33R or some younger reversed chron.

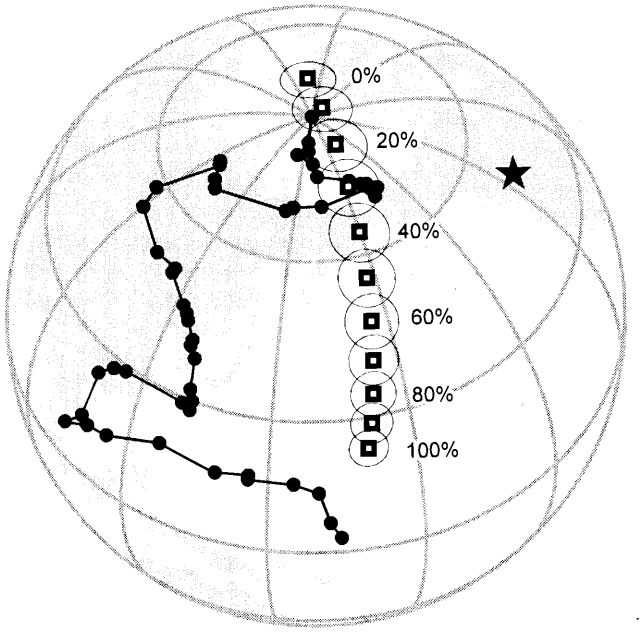


Fig. 6. North American apparent polar wander path and the position of the Crowsnest sill pole as a function of untilting. The best intersection occurs with Late Cretaceous North American poles at 30% untilting (39°).

Age of predeformational remagnetizations and local structures

High stability component remagnetizations were acquired during the Cordilleran orogeny, but prior to the local deformation. Rock magnetic characteristics, directions, and polarities all suggest that similar processes caused similar remagnetizations, asynchronously among the localities studied. The remagnetization, like the deformation, is inferred to migrate eastward, progressively anticipating the incorporation of Paleozoic strata into the accretionary prism. Within this general

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sequence of events we examine the age constraints on both the remagnetization and the deformation at individual localities.

The age of events is best constrained in the Lewis thrust sheet, where both a fold test and a contact test were performed. The pole of the remagnetization in the Lewis thrust sheet fits the North American APWP for 130 Ma and later (Barremian and later). Yet, the remagnetization occurs during a normal chron prior to internal deformation of the Lewis thrust sheet. Stratigraphic, structural, and thermochronological data suggest that motion on the Lewis thrust fault is Maastrichtian, but postdating chron 33N (earliest Maastrichtian). The youngest sediments below Lewis Thrust are early Maastrichtian (Jerzykiewicz et al. 1996). These strata are correlated to other strata deposited during chron 33N (Jerzykiewicz et al. 1996; Lerbekmo and Coulter 1985). From this we infer a maximum age for the deformation of Lewis thrust sheet that postdates chron 33N (Maastrichtian, 79–73 Ma; Harland et al. 1990). Thermal-history models constrained by apatite fission-track data from the Lewis thrust sheet indicate major cooling at about the time the youngest sediments now below the Lewis thrust fault were deposited (Maastrichtian, ~74–69 Ma; Osadetz et al. 1995). The cooling is indicative of motion on the Lewis thrust fault. Hence, thermal-history analysis provides a minimum age for the deformation that is similar to the maximum age determined from geological relationships. The remagnetization must have terminated during a normal chron prior to the deformation of the Lewis thrust sheet, likely during either the CLNS (Aptian–Santonian, 124–83 Ma) or chron 33N (Maastrichtian, 79–73 Ma). Less likely, given the pole position, would be remagnetization during M1N or M2, immediately predating the CLNS.

The reversed-polarity, syndeformational remagnetization of the Cretaceous sill in the Lewis thrust sheet occurred during a reversed chron. Following arguments made above, this event must postdate chron 33N. Therefore, the remagnetization of the sill occurred either during chron 32R (early Maastrichtian, ~73 Ma) or later, during the deformation, while the magnetic field of the Earth was reversed. Since compressive deformation ends with the onset of Eocene extension, we also assume that the remagnetization of the sill predates that event. Our preferred interpretation is that magnetization occurred at approximately 30% untilting (39°), where the apparent pole fits with the Cretaceous to Paleogene segment of the APWP (Fig. 6).

Structural and geographic position requires that the Moose Mountain culmination formed later than the Lewis thrust fault. Immediately west of the Moose Mountain culmination, Maastrichtian St. Mary River Formation lies in a syncline below the Dyson Mountain thrust (McMechan 1995). This provides a poorly constrained maximum Maastrichtian age for the structure. Nowhere, however, do stratigraphic or structural relationships place a local minimum limit on the age of the Moose Mountain structure. Likewise, the poorly constrained pole position determined for the Moose Mountain locality provides little independent constraint on the age of the local predeformational remagnetization. Following the sequence of events, at each locality, and the eastward migration of the age of the predeformational remagnetization, as inferred above, the Moose Mountain remagnetization is expected to be younger than similar remagnetizations at the Lewis locality. The reversed polarity observed at the Moose Mountain locality further limits the possible age of the remagnetization. It specifically excludes the CLNS. Since we argue that the Lewis thrust sheet remagnetization occurred in either the CLNS or 33N, we infer that the remagnetization at Moose Mountain occurred either during chron 33R (Campanian, 83–79 Ma) or later.

The observed apparent pole position for the Hosmer–Broadwood locality suggests the predeformational remagnetization was acquired during a period of normal polarity in the interval Jurassic to Neocomian, prior to the CLNS. Our data suggest that the subsequent deformation of these structures is no older than Jurassic, although it may be much younger. Other data support this inference. Structural relationships show the Hosmer–Broadwood structures are older than the Maastrichtian Lewis Thrust. Farther west, the preservation of Lower Cretaceous Blairmore Group within structures of the western ranges of the Rockies (Leech 1967) suggests that structures west of the Hosmer–Broadwood locality formed no sooner than the latest Early to Late Cretaceous. This conclusion has been debated (Price and Mountjoy 1970). Locally both the Lower Cretaceous upper Kootenay and Blairmore groups immediately east of the Hosmer Thrust contain rudaceous clasts of Late Jurassic and Early Cretaceous plutons (Norris et al. 1965). Mapped truncations of Phanerozoic strata against thrust faults in the Lizard segment of the Hosmer Nappe indicate relatively small displacements on thrusts underlying the overturned limb of the nappe (Benvenuto and Price 1979). Together these observations have been used to suggest that the location of the deformation front lay near the Hosmer Thrust in the Early Cretaceous. However, the same formations contain rudaceous clastics in the Outer Foothills (Leckie and Krystinik 1995), suggesting no simple relationship exists between clast size and proximity of the deformation front. Both the Hosmer and MacDonald thrust faults carry Upper Jurassic Fernie Group in their hanging walls and, in turn, overthrust stratigraphic successions that are conformable from the Upper Jurassic into at least Upper Cretaceous Alberta Group and possibly into Campanian Belly River Formation (Labrecque and Shaw 1973). Therefore, predeformational remagnetization of the structures in the Hosmer–Broadwood fold test occurred during a normal chron prior to the beginning of CLNS, but no older than Jurassic. The structures, themselves, are probably post-Albian, or younger, but pre-Maastrichtian in age.

Mechanism of remagnetization

We propose that a similar mechanism is responsible for the magnetization at each locality, due to the similar characteristics of the magnetization and the consistent sequence of magnetic and tectonic events. Yet, we have not determined the mechanism responsible for the predeformational remagnetization. At all three localities, the stable magnetic carrier is magnetite with similar alternating-field and thermal-unblocking spectra. Thin sections exhibit diagenetic textures that require additional study to determine whether there is a relationship among optical textures, mineral paragenesis, and the HSC of the magnetism. Other data help constrain possible mechanisms. Observed coalification and zircon fission-track data indicate that temperatures in Paleozoic strata never exceeded ~200°C (Hacquebard and Donaldson 1974;

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Osadetz et al. 1992, 1995). Such temperatures are insufficient to remagnetize magnetite by thermal reactivation alone. Thus we infer that the mechanism of predeformational remagnetization is primarily chemical. Moreover, we can infer that the mechanism is tied to orogenic processes, because of the asynchronous, eastward progression of the predeformational remagnetization in advance of the deformation.

Surprisingly, almost all remagnetization stops prior to significant deformation. It is tempting to suggest that the remagnetization is acquired immediately before or at the onset of the deformation, and that the age of the magnetic overprint dates the beginning of the local deformation. Nevertheless, there may be long intervals between the predeformational remagnetization and the local deformation.

Our inferred sequence of events does, however, predict that the age of magnetization, and its polarity should be consistent along strike. To confirm this, several localities along the length of each major fault must now be studied. Furthermore, we must broaden our stratigraphic interval of investigation. A sampling of a wider stratigraphic interval may contribute to improvements in dating the remagnetization while providing further constraints on the importance of factors such as temperature and fluid composition on chemical processes inferred to cause the remagnetization. Other data suggest that some strata in thrust sheets studied here were not remagnetized during the Cordilleran orogeny (Norris and Black 1962; Evans et al. 1975). Determination of a mechanism for the HSC predeformational remagnetization, reconciliation of other studies to our own, and a more comprehensive geographic and stratigraphic search for similar remagnetizations are important future goals for this study.

Conclusions

There is a similar, but asynchronous, sequence of events that results in an eastwardly progressing, predeformational chemical remagnetization of Paleozoic carbonate strata in the Foreland thrust and fold best of the Canadian Cordillera. A single, but unspecified and previously unrecognized, orogenic chemical process is inferred to cause the remagnetizations.

The remagnetization of western Front Ranges structures like the Hosmer Nappe and Broadwood Anticline occurred during a period of normal polarity in the Jurassic or Early Cretaceous. These structures formed after the remagnetization, but before the Maastrichtian, when motion occurred on Lewis Thrust. The predeformational remagnetization of Lewis thrust sheet Paleozoic carbonate strata occurred during either the CLNS or chron 33N, since other evidence dates motion on the Lewis Thrust as Maastrichtian. The predeformational remagnetization of the Moose Mountain culmination occurred during a reversed chron inferred no older than 33R, but possibly much younger. A syndeformational remagnetization of an Early Cretaceous sill in the Lewis thrust sheet occurred prior to the end of motion on the Lewis Thrust, when the sill had acquired 70% of its current dip, during a reversed chron no older than 32R.

This paleomagnetic study provides new insights into, and broadens the characterization of, orogenic processes, mechanisms, and timing. Our observations show that pervasive predeformational remagnetization events affected the Paleozoic carbonate succession now incorporated into the Foreland Belt of the southern Canadian Cordillera. Similarities in character and style of the remagnetizations suggest that a single chemical process of great, but hitherto unappreciated, importance was active in advance of the deformation front. This chemical process migrated progressively eastward through time in the van of the deforming Cordilleran tectonic wedge.

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